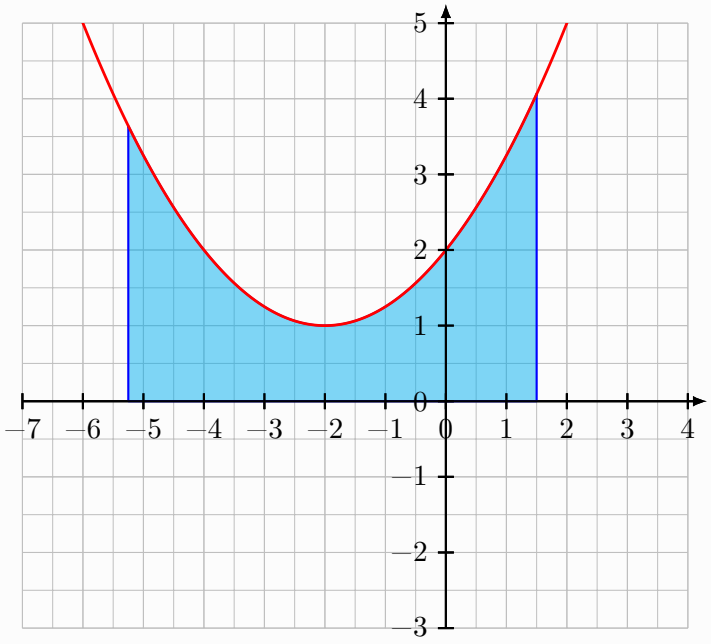
Exemples des différents cas d'intégrales (v0.32b)

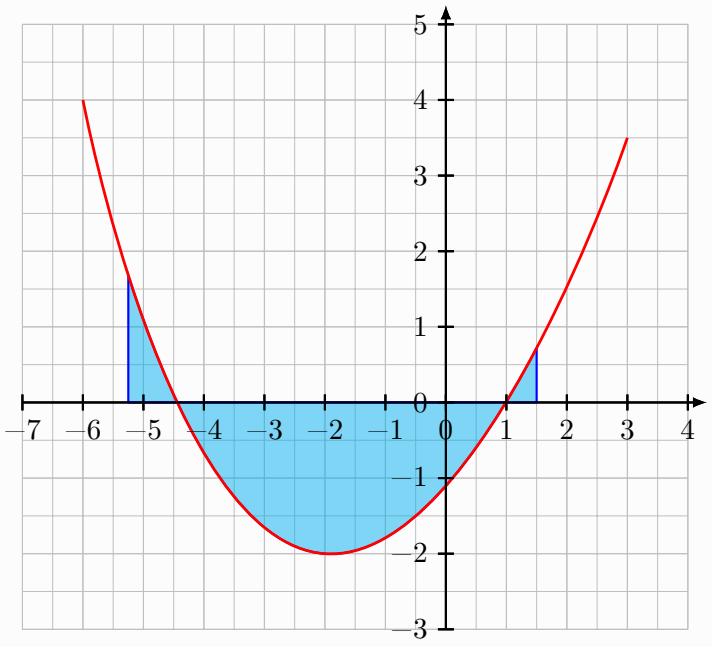
1 Sous courbe, par défaut

```
\begin{GraphiqueTikz}%
    [x=0.8cm,y=1cm,Xmin=-7,Xmax=4,Ymin=-3,Ymax=5]
  \TracerAxesGrilles[Derriere,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
  \DefinirCourbe[Nom=ch]<h>{0.25*(x+2)^2+1}
  \TracerIntegrale%
    [Couleurs=blue/cyan,Style=remplissage]%
    {h(x)} %formule
    {-5.25}{1.5}
  \TracerCourbe[Couleur=red]{h(x)}
  \TracerAxesGrilles[Devant,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
\end{GraphiqueTikz}
```



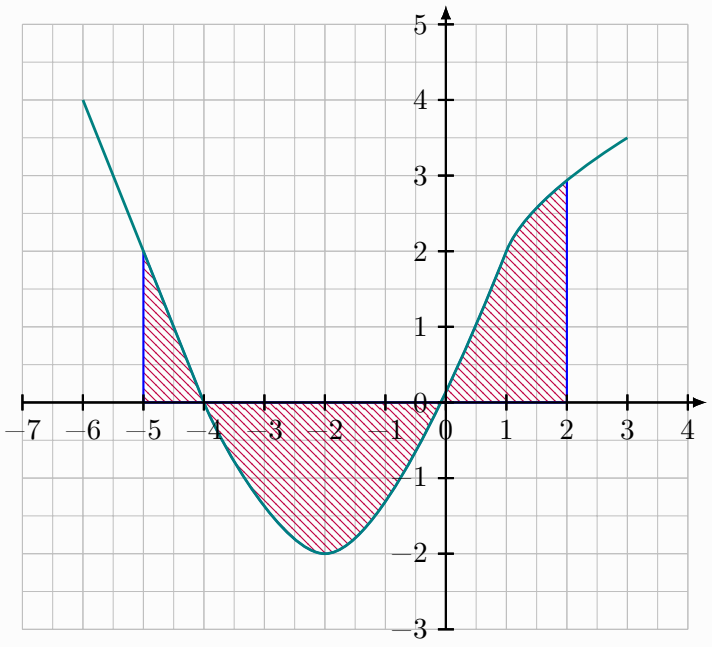
2 Sous courbe d'interpolation

```
\begin{GraphiqueTikz}%
    [x=0.8cm,y=1cm,Xmin=-7,Xmax=4,Ymin=-3,Ymax=5]
  \TracerAxesGrilles[Grads=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
  \DefinirListeInterpo{(-6,4)(-2,-2)(3,3.5)}[\interpoA]
  \DefinirCourbeInterpo[Nom=interpotest,Tension=1.05]{\interpoA}
  \TracerIntegrale%
    [NomInterpo=interpotest,Couleurs=blue/cyan,Style=remplissage,Type=itp,Tension=1.05]%
    {\interpoA} %pointsinterpo
    {-5.25}{1.5}
  \TracerCourbeInterpo[Couleur=red,Tension=1.05]{\interpoA}
  \TracerAxesGrilles[Grille=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
\end{GraphiqueTikz}
```



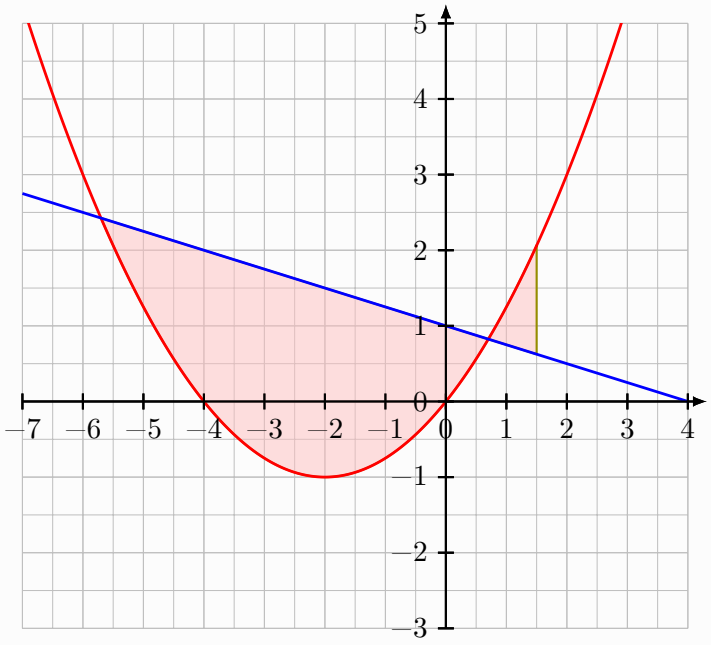
3 Sous spline cubique

```
\begin{GraphiqueTikz}%
    [x=0.8cm,y=1cm,Xmin=-7,Xmax=4,Ymin=-3,Ymax=5]
  \TracerAxesGrilles[Grads=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
  \DefinirListeSpline{-6/4/-2§-5/2/-2§-4/0/-2§-2/-2/0§1/2/2§3/3.5/0.5}[\lstsplineA]
  \DefinirCourbeSpline[Nom=splinetest]{\lstsplineA}<\SplineTeal>
  \TracerIntegrale%
    [NomSpline=\SplineTeal,Type=spl,Couleurs=blue/purple,Style=hachures]%
    {splinetest} %nomcourbespline
    {-5}{2}
  \TracerCourbeSpline[Couleur=teal]{\lstsplineA}
  \TracerAxesGrilles[Grille=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
\end{GraphiqueTikz}
```



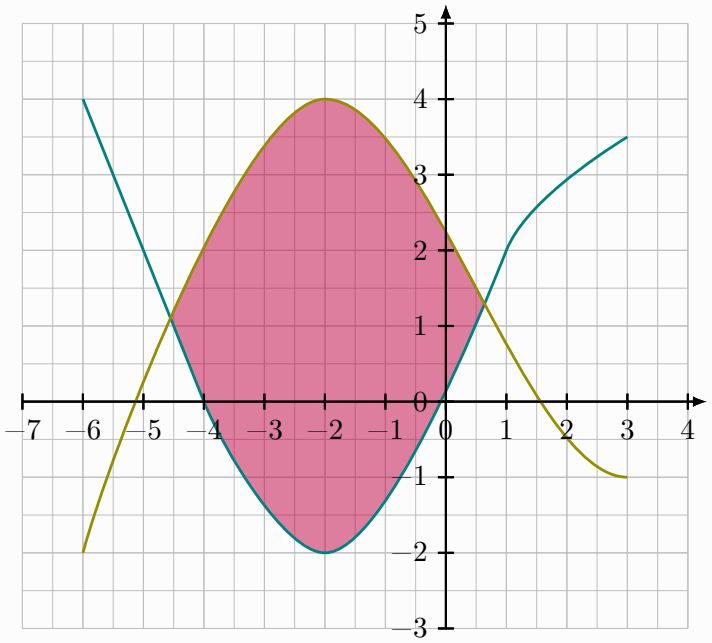
4 Entre courbes

```
\begin{GraphiqueTikz}%
  [x=0.8cm,y=1cm,Xmin=-7,Xmax=4,Ymin=-3,Ymax=5]
  \TracerAxesGrilles[Grads=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
  \DefinirCourbe[Nom=ch]<h>{0.25*(x+2)^2-1}
  \DefinirCourbe[Nom=ck]<k>{-0.25*x+1}
  \TrouverIntersections[Nom=ITSC,Aff=false]{ch}{ck}
  \TracerIntegrale%
    [Couleurs=olive/pink,Style=remplissage,Type=fct/fct,Bornes=noeud/abs]%
    {h(x)}[k(x)] %formules
    {(ITSC-1)}{1.5}
  \TracerCourbe[Couleur=red]{h(x)}
  \TracerCourbe[Couleur=blue]{k(x)}
  \TracerAxesGrilles[Grille=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
\end{GraphiqueTikz}
```



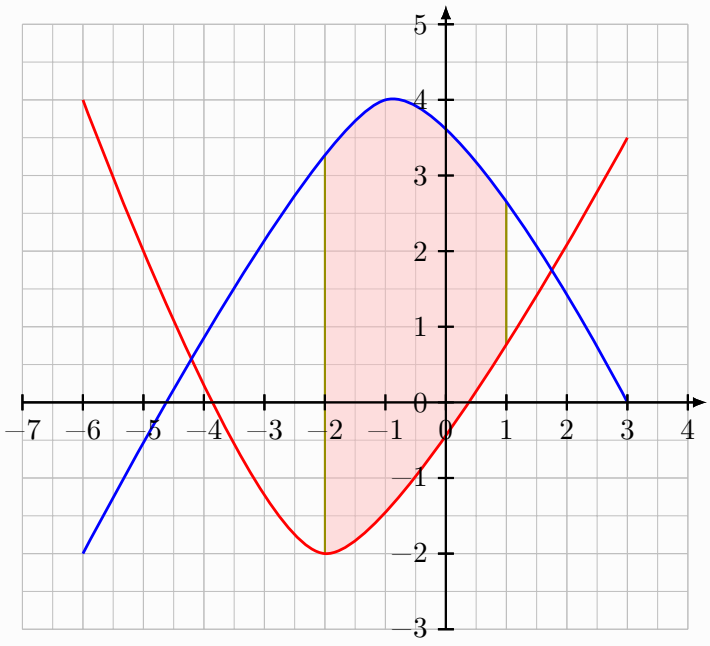
5 Entre splines

```
\begin{GraphiqueTikz}%
    [x=0.8cm,y=1cm,Xmin=-7,Xmax=4,Ymin=-3,Ymax=5]
  \TracerAxesGrilles[Grads=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
  \DefinirListeSpline{-6/4/-2§-5/2/-2§-4/0/-2§-2/-2/0§1/2/2§3/3.5/0.5}[\lstsplineA]
  \DefinirListeSpline{-6/-2/3§-2/4/0§3/-1/0}[\lstsplineB]
  \DefinirCourbeSpline[Nom=splinetestolive]{\lstsplineA}<\SplineOlive>
  \DefinirCourbeSpline[Nom=splinetestteal]{\lstsplineB}<\SplineTeal>
  \TrouverIntersections[Nom=ITT,Aff=false]{splinetestteal}{splinetestolive}
  \TracerIntegrale%
    [NomSpline=\SplineTeal,NomSplineB=\SplineOlive,Type=spl/spl,Couleurs=blue/purple,Bornes=noeuds]%
    {splinetestolive} %nomcourbespline
    [splinetestteal]  %nomcourbespline
    {(ITT-1)}{(ITT-2)}
  \TracerCourbeSpline[Couleur=teal]{\lstsplineA}
  \TracerCourbeSpline[Couleur=olive]{\lstsplineB}
  \TracerAxesGrilles[Grille=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
\end{GraphiqueTikz}
```



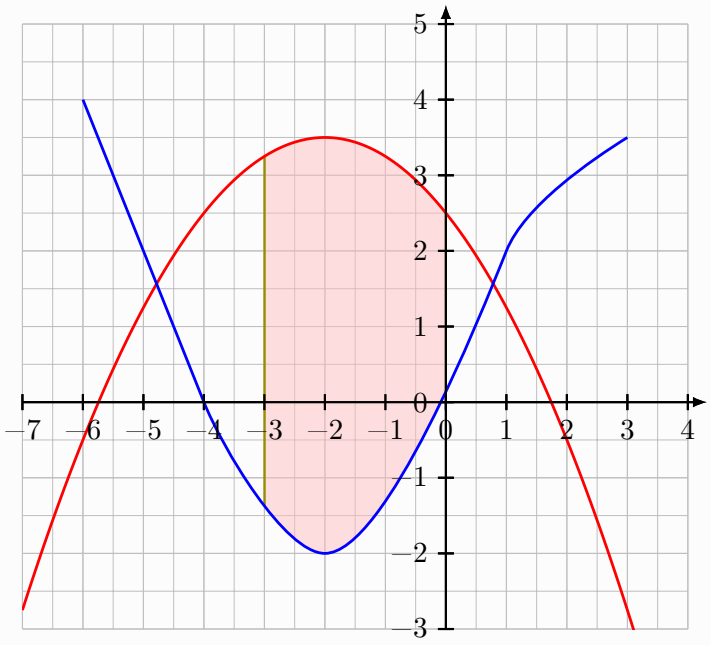
6 Entre interpolations

```
\begin{GraphiqueTikz}%
  [x=0.8cm,y=1cm,Xmin=-7,Xmax=4,Ymin=-3,Ymax=5]
  \TracerAxesGrilles[Grads=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
  \DefinirListeInterpo{(-6,4)(-2,-2)(3,3.5)}[\interpoA]
  \DefinirCourbeInterpo[Nom=interpotest]{\interpoA}
  \DefinirListeInterpo{(-6,-2)(-1,4)(3,0)}[\interpoB]
  \DefinirCourbeInterpo[Nom=interpotesta]{\interpoB}
  \TracerIntegrale%
  [NomInterpo=interpotesta,NomInterpoB=interpotest,Type=itp/itp,Couleurs=olive/pink]%
    {\interpoB} %pointsinterpo
    [\interpoA] %pointsinterpo
    {-2}{1}
  \TracerCourbeInterpo[Couleur=red]{\interpoA}
  \TracerCourbeInterpo[Couleur=blue]{\interpoB}
  \TracerAxesGrilles[Grille=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
\end{GraphiqueTikz}
```



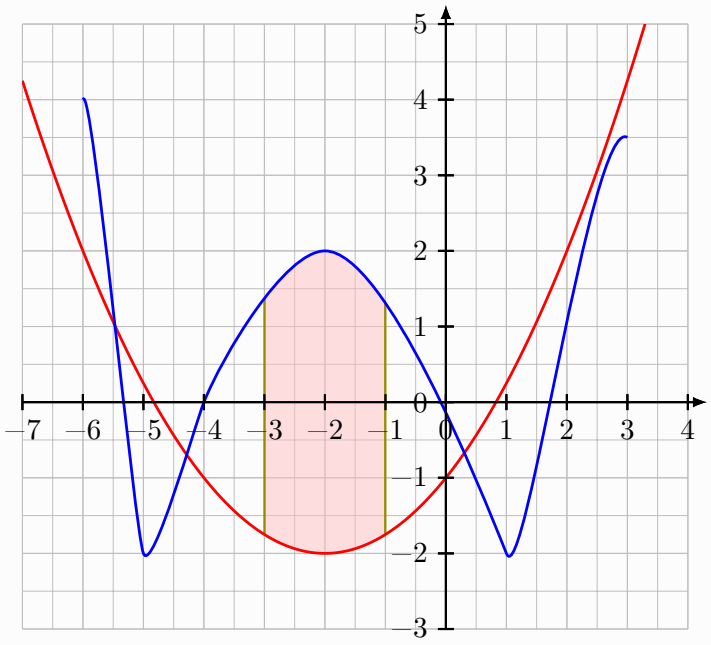
7 Entre fonction et spline

```
\begin{GraphiqueTikz}%
  [x=0.8cm,y=1cm,Xmin=-7,Xmax=4,Ymin=-3,Ymax=5]
  \TracerAxesGrilles[Grads=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
  \DefinirCourbe[Nom=ch]<h>{-0.25*(x+2)^2+3.5}
  \DefinirListeSpline{-6/4/-2§-5/2/-2§-4/0/-2§-2/-2/0§1/2/2§3/3.5/0.5}[\lstsplineA]
  \DefinirCourbeSpline[Nom=splineblue]{\lstsplineA}<\Splineblue>
  \TracerIntegrale%
    [NomSplineB=\Splineblue,Type=fct/spl,Couleurs=olive/pink]%
    {h(x)}        %formule
    [splineblue] %nomcourbespline
    {-3}{0}
  \TracerCourbe[Couleur=red]{h(x)}
  \TracerCourbeSpline[Couleur=blue]{\lstsplineA}
  \TracerAxesGrilles[Grille=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
\end{GraphiqueTikz}
```



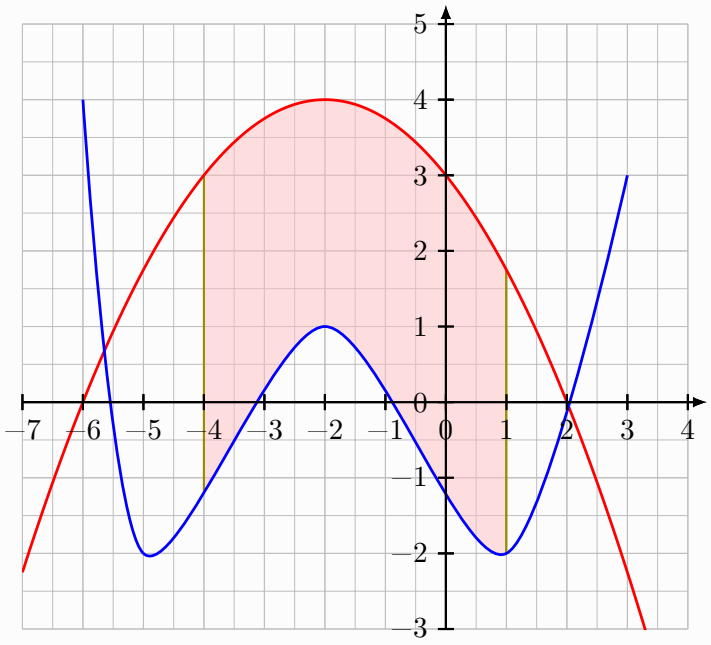
8 Entre spline et fonction

```
\begin{GraphiqueTikz}%
  [x=0.8cm,y=1cm,Xmin=-7,Xmax=4,Ymin=-3,Ymax=5]
  \TracerAxesGrilles[Grads=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
  \DefinirCourbe[Nom=ch]<h>{0.25*(x+2)^2-2}
  \DefinirListeSpline{-6/4/2§-5/-2/-2§-4/0/2§-2/2/0§1/-2/-2§3/3.5/-0.5}[\lstsplineA]
  \DefinirCourbeSpline[Nom=splineblue]{\lstsplineA}<\Splineblue>
  \TracerIntegrale%
    [NomSpline=\Splineblue,Type=spl/fct,Couleurs=olive/pink]%
    {splineblue} %nomcourbespline
    [h(x)]       %formule
    {-3}{-1}
  \TracerCourbe[Couleur=red]{h(x)}
  \TracerCourbeSpline[Couleur=blue]{\lstsplineA}
  \TracerAxesGrilles[Grille=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
\end{GraphiqueTikz}
```



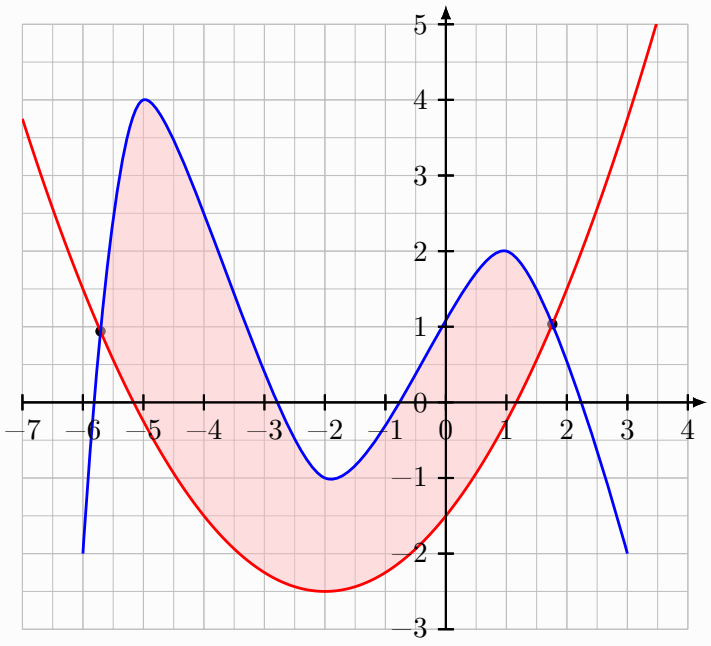
9 Entre fonction et interpo

```
\begin{GraphiqueTikz}%
  [x=0.8cm,y=1cm,Xmin=-7,Xmax=4,Ymin=-3,Ymax=5]
  \TracerAxesGrilles[Grads=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
  \DefinirCourbe[Nom=cm]<m>{-0.25*(x+2)^2+4}
  \DefinirListeInterpo{(-6,4)(-5,-2)(-2,1)(1,-2)(3,3)}[\interpoB]
  \DefinirCourbeInterpo[Nom=interpotestb]{\interpoB}
  \TracerIntegrale%
  [NomInterpoB=interpotestb,Type=fct/itp,Couleurs=olive/pink]%
    {m(x)}      %formule
    [\interpoB] %pointsinterpo
    {-4}{1}
  \TracerCourbe[Couleur=red]{m(x)}
  \TracerCourbeInterpo[Couleur=blue]{\interpoB}
  \TracerAxesGrilles[Grille=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
\end{GraphiqueTikz}
```



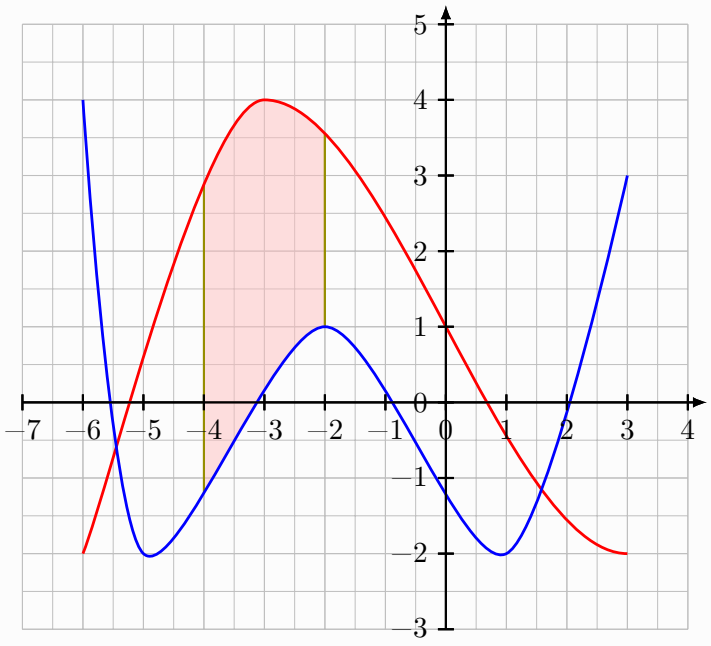
10 Entre interpo et fonction

```
\begin{GraphiqueTikz}%
  [x=0.8cm,y=1cm,Xmin=-7,Xmax=4,Ymin=-3,Ymax=5]
  \TracerAxesGrilles[Grads=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
  \DefinirCourbe[Nom=courbeQ]<q>{0.25*(x+2)^2-2.5}
  \DefinirListeInterpo{(-6,-2)(-5,4)(-2,-1)(1,2)(3,-2)}[\interpoA]
  \DefinirCourbeInterpo[Nom=interpotest]{\interpoA}
  \TrouverIntersections[Nom=FGH,Aff]{interpotest}{courbeQ}
  \TracerIntegrale%
    [NomInterpo=interpotest,Type=itp/fct,Couleurs=olive/pink,Bornes=noeuds]%
    {\interpoA} %pointsinterpo
    [q(x)]      %formule
    {(FGH-1)}{(FGH-2)}
  \TracerCourbe[Couleur=red]{q(x)}
  \TracerCourbeInterpo[Couleur=blue]{\interpoA}
  \TracerAxesGrilles[Grille=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
\end{GraphiqueTikz}
```



11 Entre spline et interpo

```
\begin{GraphiqueTikz}%
  [x=0.8cm,y=1cm,Xmin=-7,Xmax=4,Ymin=-3,Ymax=5]
  \TracerAxesGrilles[Grads=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
  \DefinirListeSpline{-6/-2/2§-3/4/0§3/-2/0}[\lstsplineA]
  \DefinirCourbeSpline[Nom=splinered]{\lstsplineA}<\Splinered>
  \DefinirListeInterpo{(-6,4)(-5,-2)(-2,1)(1,-2)(3,3)}[\interpoB]
  \DefinirCourbeInterpo[Nom=interpotestb]{\interpoB}
  \TracerIntegrale%
    [NomInterpoB=interpotestb,NomSpline=\Splinered,Type=spl/itp,Couleurs=olive/pink]%
    {splinered} %nomcourbespline
    [\interpoB] %pointsinterpo
    {-4}{-2}
  \TracerCourbeSpline[Couleur=red]{\lstsplineA}
  \TracerCourbeInterpo[Couleur=blue]{\interpoB}
  \TracerAxesGrilles[Grille=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
\end{GraphiqueTikz}
```



12 Entre interpo et spline

```
\begin{GraphiqueTikz}%
  [x=0.8cm,y=1cm,Xmin=-7,Xmax=4,Ymin=-3,Ymax=5]
  \TracerAxesGrilles[Grads=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
  \DefinirListeSpline{-6/4/-2§-3/-2.25/0§3/3/0}[\lstsplineB]
  \DefinirCourbeSpline[Nom=splineblue]{\lstsplineB}<\Splineblue>
  \DefinirListeInterpo{(-6,-2)(-5,4)(-2,-1)(1,2)(3,-2.5)}[\interpoA]
  \DefinirCourbeInterpo[Nom=interpotest]{\interpoA}
  \TrouverIntersections[Nom=UIO,Aff]{interpotest}{splineblue}
  \TracerIntegrale%
    [NomInterpo=interpotest,NomSplineB=\Splineblue,Type=itp/spl,Couleurs=olive/pink,Bornes=noeuds]%
    {\interpoA}  %pointsinterpo
    [splineblue] %nomcourbespline
    {(UIO-1)}{(UIO-2)}
  \TracerCourbeSpline[Couleur=blue]{\lstsplineB}
  \TracerCourbeInterpo[Couleur=red]{\interpoA}
  \TracerAxesGrilles[Grille=false,Elargir=2.5mm]{-7,-6,...,4}{-3,-2,...,5}
\end{GraphiqueTikz}
```

